

Brian Ponce

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EDUCATION

University of California - San Diego

Bachelor of Sciences in Mathematics-Computer Science, GPA: 3.2

La Jolla, CA

Sep. 2022 – Jun. 2026

PROJECTS

Transcript Analysis Pipeline | *Python, Jupyter, Regex, Matplotlib*

Jul. 2025

- *Built an automated text analysis tool to help fans and researchers quantify character dialogue patterns across TV series.*
- Engineered a Python pipeline to process multi-season episode transcripts, accurately attributing dialogue to characters through pattern matching and contextual analysis.
- Applied regex parsing to normalize text, capture linguistic variants, and handle edge cases such as embedded song lyrics and non-standard formatting.
- Automated frequency analysis of targeted keywords and generated statistical visualizations in Matplotlib with custom styling for enhanced readability.
- Published workflow in a Jupyter Notebook, gaining ~190 community upvotes and citation in follow-up discussions, demonstrating strong technical communication skills.

MIDI Music Generation | *Python, Jupyter, Machine Learning, Music21*

May 2025 – Jun. 2025

- *Developed algorithmic music generation system to explore computational creativity and assist composers with harmonic ideation.*
- Collaborated in a 4-person team to design end-to-end pipelines for symbolic and conditioned music generation, coordinating task distribution and integration strategies.
- Implemented a probabilistic Markov chain model for chord sequence generation, integrating it into the broader pipeline and ensuring compatibility with downstream melody generation components.
- Developed preprocessing utilities to transform Hooktheory datasets into chord/melody features for training, handling data cleaning and normalization challenges.
- Benchmarked Markov outputs against neural models, evaluating harmonic quality both quantitatively (transition probability metrics) and qualitatively (listener surveys).
- Delivered interactive demos showcasing generated music and technical workflows to peers, translating complex algorithmic concepts into accessible presentations.

Unix File Concatenation Utility | *C, File I/O, Command-Line Arguments*

Mar. 2025

- *Recreated core Unix utility to deepen understanding of systems programming and command-line tool design.*
- Implemented a functional clone of the Unix `cat` utility in C, supporting default output, line numbering (`-n` flag), and general file concatenation with proper error handling.
- Designed and tested command-line argument parsing and file I/O routines using dynamic memory and pointer handling, reinforcing low-level programming fundamentals.
- Validated correctness across multiple input scenarios including edge cases (empty files, binary data, permission errors) with custom test files.

TECHNICAL SKILLS

Languages: Python, Java, C#, C++, C

Developer Tools: Git, Vim, Visual Studio Code, Visual Studio

Libraries: pandas, NumPy, Matplotlib, re